Quantum Fault Tolerance Meets Toom's Contours

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1 Computation Model

Informally, a quantum circuit is a placement of a collection of quantum gates, taken from a finite set, in different space-time locations. Each space-time location, associated with a gate, encodes the time of the application and the qubits in the support of the gate. Formally

Separate between an element gate $g \in \mathcal{G}$ and a gate, which also has a space-time location.

Definition 1.1 (Quantum Circuit)**.** A space-time location l is a tuple that encodes in its first coordinate time, and in its second coordinate a subset of integers referring to a support of (q)bits. For a collection of quantum gates $\mathcal{G} = \{g \colon L(\mathcal{H}_2^{\otimes \Delta}) \to L(\mathcal{H}_2^{\otimes \Delta})\}$, and integers $w, d \in \mathbb{N}$ a quantum circuit C, with depth d and width w, is a collection of tuples $\{u_i = (g_i, l_i)\}_{i=1}$, where u_i are the gates of C and each u_i is composed of a gate $g_i \in \mathcal{G}$ and a space-time location $l_i \in \mathbb{Z}_d \times \mathbb{Z}_w^{\Delta}$. The gates of C have to satisfy the constraint that for every two gates with the same time coordinate, there cannot be a non-empty intersection in their space coordinate, namely for $u_i = (g_i, l_i), u_j = (g_j, l_j) \in C$, the equality $l_{i,1} = l_{j,1}$ implies $l_{i,2} \cap l_{j,2} = \emptyset$. We define the t-cut of C to be $C|_t := \{u_i = (g_i, l_i) \colon u_i \in C \text{ and } l_{i,1} = t\}$.

To define the running of a quantum circuit C on a given (mixed) quantum state, we will need to have the notion of the directed acyclic graph associated with C.

Definition 1.2 (Associated Computation DAG)**.** Let $C = \{u_i\}_{i=1}$ be a quantum circuit with depth d and width w. We define the associated computation DAG of C, denoted by π_C, to be the acyclic directed graph $\pi_C = (\mathcal{U}, E)$ as follows:

1. The vertex set $\mathcal{U}$ is the gate set of C, namely $\mathcal{U} = \{u_i\}_{i=1}$.
2. For $u_i = (g_i, l_i), u_j = (g_j, l_j)$, There is an edge $\{u_i, u_j\} \in E$ if and only if:

$$j \in \arg\min_k \left\{ l_{k,1} : (g_k, l_k) \in \mathcal{U}, l_{k,1} > l_{i,1} \text{ and } l_{k,2} \cap l_{i,2} \neq \emptyset \right\}$$

Note that there might be multiple indices that get the minimum, so the out degree might be greater than one.

In addition, suppose that $J = \{j_1, j_2, \dots, j_k\} \subset [|C|]$ is an order of the indices of the gates in C according to the topological order induced by π_C. Then, we say that J is an order of indices that agrees with π_C and define the series of circuit $C_J = \{C_{J,i}\}_{i=1}$ as follows:

1. The first element: $C_{J,1} = \{u_{j_1}\}$
2. Then, $C_{J,i+1} = C_{J,i} \cup \{u_{j_i}\}$. We name $C_{J,i}$ the partial computation circuit until step i.

Definition 1.3. Let ρ be a density matrix over w qubits, and let $C = \{u_i\}_{i=1}$ be a quantum circuit with depth d and width w. For any density matrix σ, quantum gate $g \in \mathcal{G}$, and location l, the application of the gate-location tuple (g, l) on σ, denoted by $(g, l) \circ \sigma$, is the application of g on σ where wires are ordered such that the first qubit in the input to g is $l_{2,1}$, the second is $l_{2,2}$, and so on. The result of the application of C on ρ, denoted by $C \circ \rho$, is given by iterative applications of the gates $\{u_i\}_{i=1}$ on ρ, according to a topological order induced by π_C.

We say that $\rho_1, \rho_2, \dots, \rho_k$ is a computation prefix if $\rho_1 = \rho$ and $\rho_{i+1} = u'_i \circ \rho_i$, where $\{u'_i\}_{i=1}$ is a prefix of a topological ordering of $\{u_i\}_{i=1}$. If $\{u'_i\}_{i=1}$ contains all the gates in $\{u_i\}_{i=1}$, then we call $\rho_1, \rho_2, \dots, \rho_k$ a running.

Definition 1.4 (C Subjected to $\mathcal{E}$ at frequency r)**.** Let $C = \{u_i\}_{i=1}$ be a quantum circuit with depth d and width w. For a set of fault locations $\mathcal{E} \subset [d] \times [w]$, and a number $r \in (0, 1)$ satisfies $r^{-1} \in \mathbb{N}$, we define the quantum circuit C *subjected to* $\mathcal{E}$ *at frequency* r, denoted by $C[\mathcal{E}]_r = \{u'_i\}_{i=1}$, to be the quantum circuit that is the result of the following map:

$$u_i = (g_i, (l_{i,1}, l_{i,2})) = \begin{cases} (g_i, (l_{i,1} + \lfloor r l_{i,1} \rfloor + 1, l_{i,2})) & u_i \in C \\ (g_i, (r^{-1} l_{i,1}, l_{i,2})) & u_i \in \mathcal{E} \end{cases}$$

Where r is not mentioned, we mean $r = \frac{1}{2}$, and the result of the subjection is the circuit whose gates on the odd time coordinates are the gates of C, and whose gates on the even time coordinates are the gates of $\mathcal{E}$. We will denote the associated computation DAG of C affected by $\mathcal{E}$ as $\pi[C, \mathcal{E}]$.

Definition 1.5 (Faults Propagation to a Step τ)**.** For a quantum circuit $C = \{u_i\}_{i=1}$, with depth d, an order $J = \{j_1, j_2, \dots, j_k\} \subset [|C|]$ that agrees with π_C, and gate u_i in C, a *propagation of a gate* $u_i \in C$ *to step* τ is the quantum operator $X\colon L(\mathcal{H}_2^w) \to L(\mathcal{H}_2^w)$ obtained by the follow process:

1. First, check if i precedes τ in J. If not, then return $X = I$. Otherwise, denote $j_l = i$, and look for an operator X_1 such that $u_{j_l} u_{j_{l+1}} = u_{j_{l+1}} X_1$.
2. Then, for $k \le \tau - l$, look for an operator $X_k\colon L(\mathcal{H}_2^w) \to L(\mathcal{H}_2^w)$, such that $X_{k-1} u_{j_{l+1}} = u_{j_{l+1}} X_k$.
3. At the final stage of the process, namely where $k = \tau - l$, we define $X := X_k$ to be the propagation of u_j to step τ.

We define the *propagated computation graph* $(\mathcal{U}', E')$, resulted by the propagation of u_i to step τ, as follow: Let $u_i = u_{j_l}, .., u_\tau$, be the gates at steps between i to τ, and let X be the propagation of u_i to step τ. Consider the tree rooted at u_i restricted to $u_{j_l}, .., u_\tau$, denote it by T and it leaves by L_T. Set $\mathcal{U}' = \mathcal{U}/u_i \cup X$ and:

$$\begin{aligned} E' = & E \\ & / \ \big\{\{u, u_i\} : \{u, u_i\} \in E\big\} \\ & / \ \big\{\{u, v\} : u \in L_T, \{u, v\} \in E\big\} \\ & \cup \ \big\{\{X, v\} : u \in L_T, \{u, v\} \in E\big\} \\ & \cup \ \big\{\{u, X\} : u \in L_T\big\} \\ & \cup \ \big\{\{u, v\} : \{u, u_i\}, \{u_i, v\} \in E\big\} \end{aligned}$$

For a quantum circuit C, a set of faults $\mathcal{E} = \{e_i\}_{i=1}$, an order $J \subset [|C[\mathcal{E}]|]$ that agrees with $\pi_{C[\mathcal{E}]}$, and a step $\tau \in J$, denote by $X^\tau_{e_i}$ the propagation of $e_i \in \mathcal{E}$ in $C[\mathcal{E}]$ to step τ. Let $j'_1, j'_2, \ldots, j'_l$ be the indices of the faults in $\mathcal{E}$ such that their corresponding positions in J precede τ, ordered according to the reverse topological order induced by $\pi_\mathcal{E}$. The propagated error to step τ is defined by

$$X^\tau := \prod_i X^\tau_{e_{j'_i}}$$

Definition 1.6 (Deviation)**.** For a quantum circuit C, a set of faults $\mathcal{E}$, and step τ, let X^τ be the propagated error at step τ. We define the deviation of $C[\mathcal{E}]$ from C at step τ by the set of indices on which X^τ acts non-trivially.

Definition 1.7. Using the same notations as in Definition 1.6, for a set of unitary operators $\mathcal{S} = \{s_i \colon \mathcal{H}_2^w \to \mathcal{H}_2^w\}_{i=1}$, we define the propagated error up to stabilizers in $\mathcal{S}$ as $X^\tau[S] := \min_{s \in \mathcal{S}}\{w(sX)\}$. Then, we define the deviation of $C[\mathcal{E}]$ from C at step τ up to stabilizers in $\mathcal{S}$ to be the subset of coordinates on which $X^\tau[S]$ acts non-trivially.

The following, extends the notion of Tanner graphs to the context of quantum circuits.

Definition 1.8 (Tanner Graph of a Register)**.** For a quantum circuit C, a register R is a subset of qubits in C. The *Tanner graph* at step τ of register R, denoted by $\mathcal{T}_{R,\tau}$, is the graph defined as follows:

1. If there is a code $\mathcal{C}$, such that at turn τ, the qubits in C are an encoding of $\mathcal{C}$, then $\mathcal{T}_{R,\tau}$ is the Tanner graph of $\mathcal{C}$.

2. Otherwise, $\mathcal{T}_{R,\tau}$ is the empty graph.

Where we consider a range of turns in which R is encoded in the same code, and the turn is clear from the context, we will be brief by omitting the turn subscript.

Definition 1.9 (Generalized Tanner Graph of a Circuit)**.** For a quantum circuit C, and registers a and b in C, the *generalized Tanner graph* of the register obtained by taking their union $a \cup b$ is $\mathcal{T}_{a\cup b} := \mathcal{T}_a \cup \mathcal{T}_b$. For a partition of the qubits into registers $\{R_i\}_{i=1}$, the *generalized Tanner graph* of a quantum circuit C, at step τ, according to the partition $\{R_i\}_{i=1}$, is the graph obtained by:

$$\mathcal{T}_{C,\{R_i\}_{i=1},\tau} := \bigcup_{i=1} \mathcal{T}_{R_i,\tau}$$

When the partition is clear, we will brief and use $\mathcal{T}_C$ to define the generalized Tanner graph of C.

Definition 1.10 (Clustered Deviation)**.** For a quantum circuit C, a register R of C, and a set of faults $\mathcal{E}$, let $\mathcal{T}_R$ be the Tanner graph of R, and let $\mathcal{D}_\tau$ be the deviation of $C[\mathcal{E}]$ from C on R at step τ. Denote by $\mathcal{T}_R|_{\mathcal{D}_\tau}$ the subgraph of $\mathcal{T}_R$ induced by the restriction of the left vertices (bits) to the indices in $\mathcal{D}_\tau$. Let us abuse and extend the connection notion to Tanner graphs by defining that two left vertices are connected if and only if they share a right neighbor. We define the deviation clusters at step τ to be the connected components of $\mathcal{T}_R|_{\mathcal{D}_\tau}$. The size of each deviation cluster is the number of right vertices it contains.

Claim 1.11. Let C be a quantum circuit composed of gates with bounded fan degree $\Delta \in \mathbb{N}$. In addition, let R be a quantum register encoded by a quantum error correction code $\mathcal{C}$ with Tanner graph $\mathcal{T}$ that has left and right degrees Δ_1 and Δ_2. Fix a set of faults $\mathcal{E}$, and an order J that agrees with $\pi_{C[\mathcal{E}]}$. For any τ, denote by $Y_1^{(\tau)}, Y_2^{(\tau)}, \ldots, Y_l^{(\tau)}$ the deviation clusters at step $\tau \in J$. Assume that at step $\tau_1 \in J$ there is a deviation cluster $Y_i^{(\tau_1)}$ of size $|Y_i^{(\tau_1)}|$. Then there is a deviation cluster at step $\tau_1 - 1$, let's denote it by Z, such $Z \cap Y_i^{(\tau_1)} \neq \emptyset$ and Z is of size at least:

$$|Z| \geq \frac{1}{\Delta\Delta_1\Delta_2}|Y_i^{(\tau_1)}|$$

Furthermore there is a deviation cluster at step $\tau_1 - 1$, let's denote it by Z', such $Z' \cap Y_i^{(\tau_1)} \neq \emptyset$, and Z' is of size at most:

$$|Z'| \leq \Delta\Delta_1\Delta_2|Y_i^{(\tau_1)}|$$

Add: that when it isn't mentioned, we assume that the error at time t can be decomposed into a product $\Rightarrow$ at each step, a fault is supported on at most one (q)bit.

Proof. Denote by $u = (g, l)$ the gate at step τ. If u acts trivially on the (q)bits in $Y_i^{(\tau_1)}$ then they also belong to deviation on register R at step $\tau_1 - 1$, Since they are connected in $\mathcal{T}$ they form a deviation cluster on register R at step $\tau_1 - 1$ and we got the desired. Thus it is left to handle the case in which u acts non-trivially on (q)bits in $Y_i^{(\tau_1)}$. Since u acts on at most Δ qubits, and each qubit has at most $\Delta_1(\Delta_2 - 1)$ neighbors at $\mathcal{T}$ it follows that u acts non trivially on at most

$$\Delta + \Delta\Delta_1(\Delta_2 - 1) \leq \Delta\Delta_1\Delta_2$$

deviation clusters at step $\tau_1 - 1$. Let l be integer no greater than $\Delta\Delta_1\Delta_2$ and let $Y_1, Y_2, \ldots, Y_l$ be the deviation cluster, on register R, at step $\tau_1 - 1$, on which u acts non trivially, since $Y_i^{(\tau_1)} \subset \bigcup_j Y_j$. We have that:

$$\begin{aligned} |Y_i^{(\tau_1)}| &\leq \sum_j |Y_j| \leq \Delta\Delta_1\Delta_2 \max_j |Y_j| \\ \Rightarrow \max_j |Y_j| &\geq \frac{1}{\Delta\Delta_1\Delta_2} |Y_i^{(\tau_1)}| \end{aligned} \tag{1}$$

Complete the second inequality. Should consider to erase what have already written about it. To get the second inequality, we use the fact that u is a unitary gate, in particular, u is reversible. Let $C_{J,\tau_1} = u_1, .., u_{\tau_1}$ be the partial computation circuit until step τ_1. Observe that $C_{J,\tau_1 - 1}$ is obtained by stacking $u_{\tau_1}^\dagger$ as a final stage to C_{J,τ_1}, denoted by $u^\dagger \circ C_{J,\tau_1}$. Thus, the deviation from C_{J,τ_1} at step $\tau_1 - 1$ equals the deviation from $u^\dagger \circ C_{J,\tau_1}$ at step $\tau_1 + 1$. By eq. (1), □

Corollary 1.12 (Intermediate Value Theorem)**.** Let $\alpha = 1/(\Delta\Delta_1\Delta_2)$, and let x be an integer. Assume that at the initial step τ_o, there is no deviation cluster of size more than αx, and that at a step $\tau_1 \in J$, such that $\tau_1 > \tau_o$, there is a deviation cluster $Y^{(\tau_1)}$ of size $|Y^{(\tau_1)}| > x$. Then there is a step $\tau_\star \in J$ such that $\tau_o < \tau_\star < \tau_1$ for which there is a deviation cluster Z at step $\tau_\star$, satisfying that the size of Z is in the range $(\alpha x, x)$.

Proof. Assume through contradiction that for any turn $\tau \in (\tau_o, \tau_1)$, all the deviation clusters are of size smaller than αx. Particularly, at turn $\tau_1 - 1$, all the deviation clusters are of size smaller than αx. Yet, by Claim 1.11, the existence of the deviation cluster $Y^{(\tau_1)}$, at size d, implies the existence of a deviation cluster at turn $\tau_1 - 1$, of size at least αx, which leads to a contradiction. □

Claim 1.13. Let C be a quantum circuit with order J and depth d, and let $\mathcal{E} \sim \mathcal{N}_{\mathcal{E}}$, so the circuit $C[\mathcal{E}]$ is a random variable, and it is fan-bounded by Δ. For a time $t \leq 2d$ and an integer $x \in \mathbb{N}$, let A be the event in which, until time t, there is a time $t' \leq t$ such that there is a deviation cluster of size greater than x. For any step $\tau \in J$, let B_τ be the event that step τ is the first step to have a deviation cluster of size in the range $(\alpha x, x)$. Then:

$$\mathbf{Pr}[A] \leq 2wd \max_{\tau \in [2wd]} \mathbf{Pr}[B_\tau]$$

Proof. By Corollary 1.12, we have that the occurrence of a deviation cluster of size greater than x before time t implies that there is a step $\tau \in J$, such that $\tau \leq 2wt$, in which there is a deviation cluster of size greater than αx. Thus,

$$A \subset \bigcup_{\tau \in [2wt]} B_\tau \subset \bigcup_{\tau \in [2wd]} B_\tau$$

So by using the union bound we get:

$$\begin{aligned} \mathbf{Pr}[A] \leq \mathbf{Pr}\left[\bigcup_{\tau \in [2wd]} B_\tau\right] &\leq \sum_{\tau \in [2wd]} \mathbf{Pr}[B_\tau] \\ &\leq 2wd \max_{\tau \in [2wd]} \mathbf{Pr}[B_\tau] \end{aligned}$$

□

[COMMENT] Separate the above from the beneath. Explain that we introduce two notions of propagation on purpose.

Definition 1.14 (Layer Propagation.)**.** Let $C = \{u_i\}_{i=1}$ be a quantum circuit, and let $U_1, U_2, \ldots, U_d$ be a partition of the gates in C according to their time coordination. We name such a partition the *partition of C into layers*. The *forward layer propagation* of U_i is defined to be the action that results in the quantum circuit such that its partition into layers is $U_1, \ldots, U_{i-1}, U_{i+1}, V, U_{i+2}, \ldots, U_t$, where V is a collection of quantum gates, such that

$$\prod_{u \in U_i} u \prod_{u \in U_{i+1}} u = \prod_{u \in U_{i+1}} u \prod_{u \in V} u$$

For $i' > i$, the *forward layer propagation* of U_i *to time* i' is defined by propagating U_i forward for $i' - i$ times. In the resulting circuits of the layer propagation actions, we update the time coordinates to match their application turn.

Claim 1.15. The result of forward layer propagation has a bounded fan-in.

[COMMENT] Prove it.

Definition 1.16 ($C \sim_P C'$)**.** We define the equivalence relation over quantum circuits $\sim_P$ to be satisfied if any of them can be obtained from the other by propagating layers. Namely, for quantum circuits C, C', such that the partition of C into layers is $U_1, \ldots, U_d$, we have $C \sim_P C'$ if there is a subset of layers $\{U_i\}_{i=1} \subset \{U_1, \ldots, U_d\}$ and subsets of time points $\{t_i\}_{i=1}$ such that C' is the result of propagating each of the U_i to time t_i in C according to the order indexed by i.

Note that, by definition, we find that equivalence between quantum circuits by $\sim_P$ implies an equivalence of their action. More specifically, for a mixed state ρ and quantum circuits C and C' such that $C \sim_P C'$, we have:

$$C \rho C^{\dagger} = C' \rho C'^{\dagger}$$

Definition 1.17 (Noise Channel and The Local Stochastic Nosie Property)**.** A *noise channel* $\mathcal{N}_{\mathcal{E}}$ is a distribution over sets of faults, we will denote the outcome of $\mathcal{N}_{\mathcal{E}}$ by $\mathcal{E}$. For $p \in (0,1)$ we say that a noise channel $\mathcal{N}_{\mathcal{E}}$ is a *local stochastic*, with noise rate p, if the following holds: For every subset of locations $S \subset \mathbb{N} \times \mathbb{N}$, whose locations have the same time coordinate, the probability that S is covered by $\mathcal{E}$ is less than:

$$\mathbf{Pr}_{\mathcal{E} \sim \mathcal{N}_{\mathcal{E}}} [S \subset \mathcal{E}] \leq p^{|S|}$$

Observes that a quantum circuit C and a noise channel $\mathcal{N}_{\mathcal{E}}$ induce a subjection $C[\mathcal{E}]$, which is a random variable.

Claim 1.18. If $S \subset V^{(1)}$ then for any $i \in S$ there is $j \in S$ and U_{2k} such that $j \in U_{2k}$ and U_{2k} is connect to i in $\pi|_{\leq c}$.

Claim 1.19. If $S \subset V^{(1)}$, then there is subset $S' \subset \bigcup U_{2k} \cap \Lambda(S)$ at size: X.

Proof. Suppose not, since the fan-in is at most Δ, we get that size of a connected component is at most $|S'|\Delta^c < |S|$, thus there is must to be bit in S which is not connected and therefore $S \not\subset V^{(1)}$. □

Claim 1.20. Let $\mathcal{N}_{\mathcal{E}}$ be a local stochastic noise channel with noise rate p, let C be a quantum circuit with bounded fan-in Δ, and let c be an integer. We define q to be the number that depends on p, Δ, and c as: $q := (\Delta^c p)^{\frac{1}{\Delta^c}}$. There is a local stochastic noise channel $\mathcal{N}_{\mathcal{E}_q}$, with noise rate q that induces a subjection $C[\mathcal{E}_q]_{\frac{1}{c}}$ such that:

$$C[\mathcal{E}]_{\frac{1}{2}} \sim_P C[\mathcal{E}_q]_{\frac{1}{c}}$$

Proof. We need to show that any subjection outcome $C[\mathcal{E}]$ can be transformed by propagation to a circuit C' that is also a result of subjecting C to a set of faults $\mathcal{E}'$ at frequency $\frac{1}{c}$, and that $\mathcal{E}'$ satisfies the local noise property with a noise rate q. Assume that c is even, and consider the layer propagations:

$$\begin{aligned} U_0, &U_1, U_2, .., U_{c-1}, U_c, U_{c+1}, .., U_c \mapsto \\ &U_0, U_1, U_3, U_5, .., U_{c-1}, V_2, V_4, .., V_{c-2}, U_c, U_{c+1}, U_{c+3}, .. \mapsto \\ &V^{(0)}, U_1, U_3, U_5, .., U_{c-1}, V^{(1)}, U_{c+1}, U_{c+3}, .. \end{aligned}$$

Where we grouped V_2, V_4, V_6, V_{c-2} together into $V^{(i)}$. Now, $\mathcal{E}' = \{V^{(i)}\}_{i=1}$, Now we argue that

$$\mathbf{Pr}\left[S \subset V^{(i)}\right] \le \mathbf{Pr}\left[\bigcup_{\substack{S' \in \Lambda(S) \\ |S'| = \nu|S|}} \left\{S' \subset \bigcup U_{2k}\right\}\right]$$

□

For that we need the follow property, for every subset of qubits S, there is subset of qubits S' at size at least $(1-\gamma)|S|$ such that for S to be deviated, all the qubits in S' have to absorb fault. Now suppose that the gate is at depth at most Δ, and the fanning is at most c_f, then we get:

$$\begin{aligned} &\mathbf{Pr}\left[S \in \mathcal{D}\right] \le (c_f^{\Delta} p)^{|S'|} \le (c_f^{\Delta} p)^{(1-\gamma)|S|} \\ \Rightarrow &p \to (c_f^{\Delta} p)^{1-\gamma} \end{aligned}$$

A trivial lower bound for $1-\gamma$ is .

$$\binom{\Lambda(S)}{|S'|} p^{|S'|} \sim 2^{|S|h(\alpha)} p^{|S'|}$$

[COMMENT] Denote by $\Lambda(S)$ the light cone of S. We need to iterate over all the minimal subsets of qubits that have to be flipped to make all the qubits in S flipped.

We bound this number from above by the binomial coefficient, and then by the entropy function. We have to add a definition of minimal subset. In addition, bring all the definitions about layer propagations here, and add a paragraph that explains that we separate the two notations on purpose, as each one of them is defined to prove a different theorem.